

PEACEABLE QUEENS ON THE 15×15 AND 16×16 TORI: THE EXACT VALUES $t(15) = 20$ AND $t(16) = 32$

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ABSTRACT. Let $t(n)$ denote the largest m such that m black and m white queens can be placed on the $n \times n$ toroidal chessboard with no black queen attacking a white queen. We prove that $t(15) = 20$ and $t(16) = 32$, determining the first two previously unknown terms of this sequence (OEIS A279405), whose exact values were known only through $n = 14$. Both lower bounds are explicit placements; the one for $n = 16$ is the $(4, 4)$ -plaid of Clinch, Drescher, Huynh and Saffidine. Both upper bounds are computer-assisted exhaustive proofs on a common plan: a queen placement is replaced by a colouring of the $4n$ toroidal lines, elementary counting identities restrict the line-set cardinalities to finitely many *size profiles* up to symmetry, and each profile is decided exactly by a finite search whose every pruning step is a necessary condition. For $n = 15$ a profile is the quadruple of line-set sizes; there are 247 of them modulo the induced dihedral action on the four line families and, in the top stratum, colour complementation — an action induced by an affine board-symmetry group of order 14,400 — and the search examined 6,074,753,568 cases. For $n = 16$ the diagonal and antidiagonal sizes must be refined by parity, because on an even torus a diagonal and an antidiagonal meet in two cells or in none; there are 342 canonical six-part profiles giving 677 oriented search jobs, and the search examined 13,163,028,768 cases. No larger placement was found in either search. The two values sharpen the parity contrast of the toroidal problem: $t(16)$ equals $16^2/8$ exactly, whereas $t(15)$ falls well short of $15^2/8$.

1. INTRODUCTION

A *peaceable* pair of queen armies is a placement of m black and m white queens on a chessboard such that no black queen attacks a white queen; queens of the same colour may attack each other freely. The problem of maximizing m goes back to Ainley [1] and was popularized through the OEIS sequence A250000 [3] for the ordinary $n \times n$ board. On the *toroidal* board the four lines through a cell wrap around, which makes the problem homogeneous: the symmetry group acts transitively on cells, and the combinatorics becomes arithmetic in $\mathbb{Z}/n\mathbb{Z}$. The toroidal maxima form OEIS sequence A279405 [4].

n	8	9	10	11	12	13	14	15	16
$t(n)$	8	7	12	10	18	16	25	20	32
$n^2/8$	8.0	10.1	12.5	15.1	18.0	21.1	24.5	28.1	32.0

TABLE 1. The upper end of A279405. The values through $n = 14$ are those recorded in [4]; $t(13) = 16$ and $t(14) = 25$ were contributed by Jaideep Padhi in June 2026. The values at $n = 15$ and $n = 16$ are Theorems 1 and 2.

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Table 1 shows the characteristic parity split of the toroidal problem: for even n the known values sit at or near $n^2/8$ and at $n = 14$ exceed it, whereas for odd n they fall well short of it, hovering nearer $n^2/11$. Clinch, Drescher, Huynh and Saffidine [2] made this contrast precise in the asymptotic regime, proving $t(n) \leq 0.125n^2$ for odd n [2, Theorem 1.4] against $t(n) \geq 0.1339n^2$ for all sufficiently large even n [2, Theorem 1.1], the latter by an explicit family of constructions they call *plaids*. Their best published upper bound for the even torus is $t(n) \leq 0.1402n^2$ for all even n [2, Theorem 1.2], the numerical optimum of the associated non-linear program being $0.140132\dots$ [2, §4.2].

At $n = 15$ and $n = 16$ this left two gaps. For $n = 15$, Theorem 1.4 of [2] gives $t(15) \leq 28$ by integrality, and the appendix of [2] exhibits an explicit 20+20 configuration, so $t(15) \in [20, 28]$. For $n = 16$, Theorem 1.2 of [2] gives $t(16) \leq 35$ by integrality, and the $(4, 4)$ -plaid on the 16×16 torus is peaceable with 32 queens of each colour (Section 7), so $t(16) \in [32, 35]$. A279405 listed no term for either n .

We determine both.

Theorem 1. *On the 15×15 toroidal chessboard, 20 black and 20 white queens can be placed with no black–white attack, and 21 black and 21 white queens cannot. That is, $t(15) = 20$.*

Theorem 2. *On the 16×16 toroidal chessboard, 32 black and 32 white queens can be placed with no black–white attack, and 33 black and 33 white queens cannot. That is, $t(16) = 32$.*

Both lower bounds are placements, reproduced in Sections 3 and 7; neither is new. The upper bounds are the substance of this note and are computer-assisted. They follow a common plan, and the two cases differ in exactly the place where the arithmetic of $\mathbb{Z}/n\mathbb{Z}$ differs.

Section 2 eliminates the queens, for every n at once. A peaceable $m + m$ placement exists if and only if the $4n$ lines of the board can be coloured black and white so that the set of cells all four of whose lines are black, and the set of cells all four of whose lines are white, both have at least m elements. The black line sets $R, C, D, A \subseteq \mathbb{Z}/n\mathbb{Z}$ (rows, columns, diagonals, antidiagonals) then determine everything, and swapping colours if necessary one may assume $S := |R| + |C| + |D| + |A| \leq 2n$. We stress that $S \leq 2n$ is genuinely an inequality: it cannot be normalized to $S = 2n$, because adding a line to the black side removes it from the white side and may destroy the white count. Section 2 also records the two counting facts on which everything rests: which pairs of lines meet in how many cells, and an exact identity for triples of families.

The single arithmetic difference between the two cases is the incidence of a diagonal with an antidiagonal. The cells on diagonal d and antidiagonal a are the solutions of $2r = a + d$, $2c = a - d$. For odd n the element 2 is a unit, so there is exactly one such cell, and the four line families behave symmetrically. For even n there are two such cells when $d \equiv a \pmod{2}$ and none otherwise. This costs the even case one symmetry — the halving map that exchanges $\{R, C\}$ with $\{D, A\}$ is no longer available — and gains it a refinement: the diagonal and antidiagonal sizes must be split by the parity of the line label.

Accordingly, Sections 3–6 treat $n = 15$ and Sections 7–10 treat $n = 16$, in parallel. In each case one bounds the possible size profiles by counting (Sections 4, 8), decides a single profile by an exact three-stage search in which every pruning rule is a necessary condition (Sections 5, 9), and reports the outcome (Sections 6, 10): 247 canonical profiles and 6,074,753,568 cases for $n = 15$, all infeasible; 342 canonical profiles, 677 oriented jobs and 13,163,028,768 cases for $n = 16$, all infeasible.

Section 11 describes the verification, which deserves the most scrutiny and accordingly receives the most space, with a subsection for each case. The evidence is not equally strong on the two sides, and we say so there: for $n = 15$ two independently written enumerators with

different symmetry quotients and different completion algorithms agree on all 247 profiles, whereas for $n = 16$ the two programs that were run share everything except the completion kernel. Sections 12 and 13 give reproduction instructions and disclose the use of AI systems in producing this note.

Remark 3. Both proofs are exhaustive rather than explanatory. They give no human-readable reason why the odd torus is so much poorer than the even torus, and they do not sharpen the general bounds of [2]. A structural proof of $t(15) \leq 20$ — or, better, of a general odd- n bound near $n^2/11$ — remains open and would supersede the first computation; likewise a structural proof that the plaid construction is optimal on the even torus would supersede the second.

Remark 4. Conjecture 5.1 of [2] predicts $t(n) \leq 2 + \max \min\{|B|, |W|\}$ for even n , the maximum taken over (a, b) -plaids on the $n \times n$ torus. A direct check of the 64 plaids available at $n = 16$ gives $\max \min\{|B|, |W|\} = 32$, so the conjecture predicts $t(16) \leq 34$. Theorem 2 confirms the conjecture at $n = 16$ and shows that its slack of 2 is not attained there.

2. NOTATION AND THE LINE-COLOURING REDUCTION

Throughout, $n \geq 2$ and all coordinates lie in $\mathbb{Z}/n\mathbb{Z}$, represented by $\{0, 1, \dots, n-1\}$. The board is $G = (\mathbb{Z}/n\mathbb{Z})^2$, with n^2 cells. Through a cell (r, c) pass four *lines*:

$$\text{row } r, \quad \text{column } c, \quad \text{diagonal } d = r - c, \quad \text{antidiagonal } a = r + c,$$

each family indexed by $\mathbb{Z}/n\mathbb{Z}$, so the board carries $4n$ lines, each containing exactly n cells. Two queens *attack* each other exactly when they share a line. We write $t(n)$ for the largest m such that m black and m white queens can be placed on the $n \times n$ torus with no black–white attack. All $2m$ queens occupy distinct cells; queens of one colour may share lines.

For $R, C, D, A \subseteq \mathbb{Z}/n\mathbb{Z}$ put

$$B(R, C, D, A) = \{(r, c) \in G : r \in R, c \in C, r - c \in D, r + c \in A\},$$

and let $W(R, C, D, A) = B(R^c, C^c, D^c, A^c)$, where c denotes complement in $\mathbb{Z}/n\mathbb{Z}$ taken family by family. Cells of B are called *black-homogeneous* and cells of W *white-homogeneous*.

Lemma 5 (Line-colouring equivalence). *Let $m \geq 1$. There is a peaceable placement of m black and m white queens on the $n \times n$ torus if and only if there are sets $R, C, D, A \subseteq \mathbb{Z}/n\mathbb{Z}$ with*

$$|B(R, C, D, A)| \geq m \quad \text{and} \quad |W(R, C, D, A)| \geq m.$$

Proof. ($\Rightarrow$) Given a peaceable placement, colour every line carrying a black queen black and every line carrying a white queen white. No line receives both colours: a line carrying queens of both colours would exhibit a black–white attack. Colour every remaining line white, and let R, C, D, A be the black line sets in the four families; the white line sets are then exactly the complements R^c, C^c, D^c, A^c . Every black queen sits on four black lines, so its cell lies in B , and the m black queens occupy distinct cells, whence $|B| \geq m$; likewise every white queen sits on four lines that carry no black queen, hence on four white lines, so $|W| \geq m$.

($\Leftarrow$) Given such sets, place black queens on any m cells of B and white queens on any m cells of W . The two sets are disjoint, since a cell cannot have all four of its lines both in and out of the black sets. If a black queen attacked a white queen they would share a line, say a row r ; but then $r \in R$ and $r \in R^c$, which is absurd. $\square$

Colouring the unused lines white rather than leaving them unused is harmless in the direction that matters: it can only enlarge the white sets, and W is monotone increasing in the complements. From now on, a *configuration* is a quadruple (R, C, D, A) of subsets of $\mathbb{Z}/n\mathbb{Z}$. Write

$$s_0 = |R|, \quad s_1 = |C|, \quad s_2 = |D|, \quad s_3 = |A|, \quad S = s_0 + s_1 + s_2 + s_3.$$

Lemma 6 (Budget). *If some configuration has $|B| \geq m$ and $|W| \geq m$, then some configuration with the same property has $S \leq 2n$.*

Proof. Replacing (R, C, D, A) by (R^c, C^c, D^c, A^c) exchanges B and W , hence preserves the property, and replaces S by $4n - S$. One of S and $4n - S$ is at most $2n$. $\square$

Remark 7. Lemma 6 is used only as an inequality. We do not normalize to $S = 2n$: no lemma asserting that a feasible configuration can always be modified into one with $S = 2n$ is proved or used here, and none is obviously true, because feasibility is not monotone under transferring a line from the white side to the black side — a line added to R is removed from R^c and can destroy $|W| \geq m$. (It can also fail to: the witness of Proposition 10, with black-side $S = 28$, happens to extend to a configuration with $S = 30$ and $|B| = |W| = 20$ by adding row 2 and column 11, so individual examples decide nothing either way.) Accordingly both searches below conservatively cover the whole admissible range of S , not merely the top stratum. For $n = 16$ the necessary conditions of Section 8 turn out to admit no profile at all with $S \in \{30, 31, 32\}$.

Lemma 8 (Incidence). *Let X and Y be two distinct line families and let ℓ be a line of X and ℓ' a line of Y .*

- (i) *If $\{X, Y\} \neq \{D\text{-family}, A\text{-family}\}$, then ℓ and ℓ' meet in exactly one cell.*
- (ii) *If ℓ is the diagonal d and ℓ' the antidiagonal a , their common cells are the solutions of*

$$(1) \quad 2r = a + d, \quad 2c = a - d \quad \text{in } \mathbb{Z}/n\mathbb{Z}.$$

For odd n there is exactly one. For even n there are exactly two if $d \equiv a \pmod{2}$, and none otherwise; the two are (r, c) and $(r + \frac{n}{2}, c + \frac{n}{2})$.

Proof. (i) A row r and a column c meet in (r, c) ; a row r and a diagonal d in $(r, r - d)$; a row r and an antidiagonal a in $(r, a - r)$; a column c and a diagonal d in $(c + d, c)$; a column c and an antidiagonal a in $(a - c, c)$. In each case the cell is unique.

(ii) A cell (r, c) lies on diagonal d and antidiagonal a iff $r - c = d$ and $r + c = a$; adding and subtracting gives (1), and conversely (1) together with either original equation gives the other. Consider the homomorphism $\varphi: G \rightarrow (\mathbb{Z}/n\mathbb{Z})^2$, $\varphi(r, c) = (r - c, r + c)$. Its kernel is $\{(r, c) : r = c, 2r = 0\}$. For odd n the kernel is trivial and φ is a bijection, with inverse $r = 2^{-1}(d + a)$, $c = 2^{-1}(a - d)$; so each fibre is a single cell. For even n the kernel is $\{(0, 0), (\frac{n}{2}, \frac{n}{2})\}$, of order 2, so every nonempty fibre has exactly two elements and the image has $n^2/2$ elements. Every (d, a) in the image satisfies $d + a = 2r \equiv 0 \pmod{2}$, and the set of such pairs also has $n^2/2$ elements; hence the image is exactly $\{(d, a) : d \equiv a \pmod{2}\}$. The description of the second solution is the kernel translate. $\square$

For two families X, Y with black sets $\mathcal{X}, \mathcal{Y}$ write $m(X, Y)$ for the number of cells lying on a black line of X and on a black line of Y . By Lemma 8, $m(X, Y) = |\mathcal{X}| |\mathcal{Y}|$ unless $\{X, Y\}$ is the diagonal–antidiagonal pair; the exceptional value is computed in each case below.

Lemma 9 (Triple identity). *Let X, Y, Z be three of the four families, with black sets of sizes x, y, z . Let T be the number of cells whose lines in these three families are all black, and T' the number whose lines in these three families are all white. Then*

$$(2) \quad T + T' = n^2 - n(x + y + z) + m(X, Y) + m(X, Z) + m(Y, Z).$$

Consequently, if $|B| \geq m$ and $|W| \geq m$ then the right-hand side of (2) is at least $2m$, for each of the four triples of families.

Proof. Write $u, v, w: G \rightarrow \{0, 1\}$ for the indicators of “the cell’s line in family X (resp. Y , Z) is black”. Then

$$uvw + (1 - u)(1 - v)(1 - w) = 1 - u - v - w + uv + uw + vw.$$

Summing over the n^2 cells: $\sum 1 = n^2$; $\sum u = nx$, since each of the x black lines of family X contains n cells, and likewise $\sum v = ny$, $\sum w = nz$; and $\sum uv = m(X, Y)$ by definition, likewise for the other two pairs. This is (2). Every cell of B is counted by T and every cell of W by T' , and the two sets are disjoint, so $|B| + |W| \leq T + T'$. $\square$

Part I: the odd case $n = 15$.

3. THE LOWER BOUND FOR $n = 15$

Proposition 10. $t(15) \geq 20$.

Proof. Take the black line sets

$$(3) \quad \begin{aligned} R &= \{5, 7, 8, 11, 12, 14\}, & D &= \{0, 1, 4, 8, 12, 13, 14\}, \\ C &= \{0, 3, 4, 7, 9, 10, 12, 13\}, & A &= \{1, 2, 3, 9, 10, 12, 14\}, \end{aligned}$$

of sizes $(6, 8, 7, 7)$ in $\mathbb{Z}/15\mathbb{Z}$, and colour every remaining line white, giving white line sets of sizes $(9, 7, 8, 8)$. Direct enumeration of the 225 cells gives exactly 20 black-homogeneous cells,

$$\begin{array}{cccc} (5, 4) & (5, 7) & (5, 12) & \\ (7, 3) & (7, 7) & (7, 9) & (7, 10) \\ (8, 4) & (8, 9) & (8, 10) & \\ (11, 3) & (11, 7) & (11, 13) & \\ (12, 0) & (12, 4) & (12, 12) & (12, 13) \\ (14, 0) & (14, 10) & (14, 13) & \end{array}$$

and exactly 22 white-homogeneous cells, of which we use the following 20:

$$\begin{array}{ccc} (0, 5) & (0, 6) & (0, 8) \\ (1, 5) & (1, 6) & (1, 14) \\ (2, 6) & (2, 11) & \\ (3, 1) & (3, 8) & \\ (4, 1) & (4, 2) & \\ (6, 1) & (6, 14) & \\ (9, 2) & (9, 6) & (9, 14) \\ (10, 1) & (10, 5) & (13, 2) \end{array}$$

(the two unused ones are $(13, 6)$ and $(13, 8)$). Place black queens on the first list and white queens on the second. Every black queen lies on four black lines and every white queen on four white lines; a black–white attack would require a line that is both black and white, and there is none. Hence 20 black and 20 white queens coexist peaceably. $\square$

$r \setminus c$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0	.	.	.	.	.	□	□	.	□	.	.	.	.	.	.
1	.	.	.	.	.	□	□	.	.	.	.	.	.	.	□
2	.	.	.	.	.	.	□	.	.	.	.	□	.	.	.
3	.	□	.	.	.	.	.	.	□	.	.	.	.	.	.
4	.	□	□	.	.	.	.	.	.	.	.	.	.	.	.
5	.	.	.	.	■	.	.	■	.	.	.	.	■	.	.
6	.	□	.	.	.	.	.	.	.	.	.	.	.	.	□
7	.	.	.	■	.	.	.	■	.	■	■	.	.	.	.
8	.	.	.	.	■	.	.	.	.	■	■	.	.	.	.
9	.	.	□	.	.	.	□	.	.	.	.	.	.	.	□
10	.	□	.	.	.	□	.	.	.	.	.	.	.	.	.
11	.	.	.	■	.	.	.	■	.	.	.	.	.	■	.
12	■	.	.	.	■	.	.	.	.	.	.	.	■	■	.
13	.	.	□	.	.	.	◇	.	◇	.	.	.	.	.	.
14	■	.	.	.	.	.	.	.	.	.	■	.	.	■	.

FIGURE 1. The $20+20$ placement of Proposition 10: ■ black queens, □ white queens, ◇ the two white-homogeneous cells left empty. Rows, columns and both diagonal families wrap around.

Figure 1 displays the configuration. This lower bound is not new: an explicit $20+20$ configuration for $n=15$ is given in the appendix of [2]. We reproduce a placement here in line-set form so that Theorem 1 is self-contained and checkable by hand.

4. ADMISSIBLE SIZE PROFILES FOR $n=15$

Here $n=15$, 2 is a unit modulo 15 with $2^{-1}=8$, and by Lemma 8 any two lines from distinct families meet in exactly one cell; in particular $m(X, Y) = |\mathcal{X}||\mathcal{Y}|$ for all six pairs. We call (s_0, s_1, s_2, s_3) the *size profile* of the configuration and ask whether a configuration exists with $|\mathcal{B}| \geq 21$ and $|\mathcal{W}| \geq 21$.

Lemma 11 (Pair bounds). *Let (R, C, D, A) be a configuration in $\mathbb{Z}/15\mathbb{Z}$ with size profile (s_0, s_1, s_2, s_3) . Then for all $i < j$,*

$$|\mathcal{B}| \leq s_i s_j, \quad |\mathcal{W}| \leq (15 - s_i)(15 - s_j).$$

Consequently, if $|\mathcal{B}| \geq 21$ and $|\mathcal{W}| \geq 21$ then

$$(4) \quad s_i s_j \geq 21 \quad \text{and} \quad (15 - s_i)(15 - s_j) \geq 21 \quad \text{for all } i < j.$$

Proof. Every cell of $\mathcal{B}$ lies on a black line of family i and on a black line of family j , and distinct cells of $\mathcal{B}$ give distinct such pairs of lines, because by Lemma 8(i)–(ii) two lines from distinct families meet in exactly one cell when n is odd; hence $|\mathcal{B}| \leq s_i s_j$. The bound for $\mathcal{W}$ is the same statement applied to the complementary sets. $\square$

By Lemma 9 with $m(X, Y) = xy$ throughout, if $|\mathcal{B}| \geq 21$ and $|\mathcal{W}| \geq 21$ then

$$(5) \quad 225 - 15(x + y + z) + xy + xz + yz \geq 42$$

for each of the four triples of families, where x, y, z are the corresponding sizes. Note that (5) is symmetric under $(x, y, z) \mapsto (15 - x, 15 - y, 15 - z)$, as it must be.

Symmetry. Let Γ be the group of affine bijections of G that permute the four line families. It consists of the 225 translations composed with 64 linear maps, so $|\Gamma| = 14,400$; this was verified by exhaustive enumeration of the invertible 2×2 matrices over $\mathbb{Z}/15\mathbb{Z}$. Every $\gamma \in \Gamma$ carries a configuration (R, C, D, A) to a configuration with the same $|B|$ and $|W|$, permuting the families according to a homomorphism $\Gamma \rightarrow \text{Sym}\{R, C, D, A\}$. The image is the dihedral group of order 8 preserving the partition $\{\{R, C\}, \{D, A\}\}$, generated by

$$\begin{aligned} \text{transpose } \tau(r, c) = (c, r) : & \quad R \leftrightarrow C, \ D \mapsto -D, \ A \mapsto A; \\ \text{reflection } \nu(r, c) = (r, -c) : & \quad R \mapsto R, \ C \mapsto -C, \ D \leftrightarrow A; \\ \text{halving } \psi(r, c) = (r + c, r - c) : & \quad \{R, C\} \leftrightarrow \{D, A\}. \end{aligned}$$

For ψ one checks directly that a level set of the new row coordinate is an old antidiagonal, of the new column coordinate an old diagonal, of the new diagonal an old column and of the new antidiagonal an old row; ψ is bijective because $\det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -2$ is a unit modulo 15.

Composing τ , ν and ψ yields, in one-line notation listing the images of (R, C, D, A) , exactly the eight family permutations

$$RCDA, \quad RCAD, \quad CRDA, \quad CRAD, \quad DARC, \quad DACR, \quad ADRC, \quad ADCR,$$

which one verifies by direct substitution to be closed under composition: they form the dihedral group of order 8 preserving the partition $\{\{R, C\}, \{D, A\}\}$, and no composition of the three generators produces a permutation splitting that partition. Conversely, every family-preserving linear map agrees with one of eight representatives — one above each induced permutation — up to composition with a common unit scaling $(r, c) \mapsto (ur, uc)$, and $|(\mathbb{Z}/15\mathbb{Z})^\times| = 8$, which accounts for all $8 \cdot 8 = 64$ linear maps found by the exhaustive matrix enumeration. The induced action on size profiles is the permutation action of this D_4 on ordered quadruples.

Proposition 12. *Let (R, C, D, A) be a configuration in $\mathbb{Z}/15\mathbb{Z}$ with $|B| \geq 21$, $|W| \geq 21$ and $S \leq 30$. Then its size profile is equivalent, under the D_4 action together with complementation $s_i \mapsto 15 - s_i$ in the case $S = 30$, to one of exactly 247 canonical profiles, distributed by S as in Table 2.*

S	20	21	22	23	24	25	26	27	28	29	30
canonical profiles	1	1	4	6	13	18	29	37	47	52	39

TABLE 2. The 247 canonical size profiles for $n = 15$, by budget S .

Proof. Filter all 16^4 quadruples $(s_0, s_1, s_2, s_3) \in \{0, \dots, 15\}^4$ by $S \leq 30$ and by conditions (4) and (5), which are necessary by Lemmas 11 and 9, and canonicalize the survivors: replace each by the lexicographically least element of its D_4 orbit, adjoining the D_4 orbit of the complementary profile $(15 - s_i)_i$ when $S = 30$. Complementation preserves the property “ $|B| \geq 21$ and $|W| \geq 21$ ” but sends S to $60 - S$, so it may be used as an identification only within the stratum $S = 30$, which it preserves. The enumeration is a finite computation over 65,536 quadruples; it was carried out independently three times, in two languages, always giving the same 247 profiles and the distribution of Table 2. $\square$

Inspection of the list shows that every surviving profile satisfies $3 \leq s_i \leq 12$ for all i ; the unique profiles at the two smallest budgets are $(5, 5, 5, 5)$ and $(5, 5, 5, 6)$, while the top stratum $S = 30$ contains both the lopsided $(3, 8, 9, 10)$ and the balanced $(7, 7, 8, 8)$.

5. DECIDING ONE PROFILE EXACTLY FOR $n = 15$

Fix a canonical profile. By the D_4 symmetry we may choose either (R, C) or (D, A) as the *fixed pair*; write (r, c) for its two sizes, with $r \leq c$ after a transpose if necessary, and (d, a) for the sizes of the two remaining families, which we call the *completion* families. The search proceeds in three nested stages: orbit representatives for the fixed pair, all subsets for one completion family, and an exact decision for the last.

Stage 1: representatives of the fixed pair. Fix the pair (R, C) up to a subgroup $H \leq \Gamma$ of the stabilizer of the ordered profile that also preserves the chosen fixed-pair/completion decomposition. It is generated by:

- independent translations $(R, C) \mapsto (R + \alpha, C + \beta)$, which change D and A only by translations (225 elements);
- common unit scaling $(R, C) \mapsto (uR, uC)$ for $u \in (\mathbb{Z}/15\mathbb{Z})^\times$, which scales D and A by the same u (8 elements);
- the transpose τ , admissible exactly when $r = c$, since τ exchanges the two fixed families while sending $D \mapsto -D$ and fixing A , hence preserving the completion sizes;
- the relative sign ν , i.e. $(R, C) \mapsto (R, -C)$, admissible *only* when $d = a$.

The last restriction is the one soundness point that is easy to get wrong. Since ν exchanges the diagonal and antidiagonal families, an ordered profile with $d \neq a$ is not mapped to itself, and quotienting by ν would then identify configurations belonging to different ordered profiles — legitimate only if both orientations are enumerated. We use ν only when $d = a$, where it is a genuine automorphism of the ordered profile. An earlier cost model in this project applied ν unconditionally; correcting it raised the workload of the independent enumerator from 4,876,723,839 to 6,241,793,402 completions.

H need not be the full stabilizer of the ordered profile inside Γ : when all four sizes are equal, for instance, ψ stabilizes the ordered profile but exchanges the fixed pair with the completion pair, and it is deliberately not in H . Quotienting by a subgroup is always sound — it can only produce more representatives than a larger group would, never miss an orbit — so omitted symmetries cost work, not correctness. The quotient is justified by the following transfer lemma.

Lemma 13 (Orbit transfer). *Let (R, C) and (R', C') lie in the same H -orbit. Then for each pair of prescribed completion sizes (d, a) , a completion (D, A) of (R, C) with $|B| \geq 21$, $|W| \geq 21$ and $|D| = d$, $|A| = a$ exists if and only if the corresponding completion of (R', C') exists.*

Proof. An element of H with $(R, C) = h(R', C')$ is induced by an affine board symmetry $\gamma \in \Gamma$. If (R', C', D', A') is a configuration, then γ carries it to a configuration with fixed pair (R, C) and the same $|B|$ and $|W|$, and by the admissibility conditions above (τ only when $r = c$, ν only when $d = a$) the completion sizes are preserved. Apply this to a feasible completion in either direction. $\square$

By Lemma 13 it suffices that the enumerated set of pairs intersect every H -orbit of pairs of the prescribed sizes; nothing more is required for soundness. The group has order $225 \cdot 8 = 1800$, doubled for each of the two admissible extra generators, hence 1800, 3600 or 7200. The

implementation produces its pair set by reducing each coordinate to its cyclic-translation necklace and then sweeping all necklace pairs: each pair not previously marked is emitted as a representative, and all images of an emitted pair under the residual unit, transpose and sign actions are marked. A pair is skipped only when marked, i.e. only when an explicit $h \in H$ mapping an emitted representative to it has been exhibited, so the emitted set meets every orbit. That it is moreover an exact transversal — one representative per orbit — is not needed for soundness but is verified anyway: only 43 distinct triples (r, c, group) occur across the 247 profiles, and for each of them the cardinality of the emitted set equals the orbit count computed independently by Burnside’s lemma (Section 11).

Choosing the fixed pair is a free optimization: for each profile both choices are costed as (number of pair representatives) $\times$ (number of subsets of the smaller completion family) and the cheaper one is taken. Likewise, within the completion families, the family with fewer subsets is the one enumerated in Stage 2, which is legitimate because ν exchanges the two completion families globally, so the ordered profiles (r, c, d, a) and (r, c, a, d) are simultaneously feasible or infeasible.

Stage 2: the completion as a subset problem. Fix a representative (R, C) . Index cells by their diagonal and antidiagonal coordinates via the inverse $r = 8(d + a)$, $c = 8(a - d)$ of Lemma 8(ii), writing x for the diagonal and y for the antidiagonal coordinate to keep them apart from the sizes d, a , and set

$$P_{x,y} = \mathbf{1}_R(8(x+y)) \mathbf{1}_C(8(y-x)), \quad Q_{x,y} = \mathbf{1}_{R^c}(8(x+y)) \mathbf{1}_{C^c}(8(y-x)) \quad (x, y \in \mathbb{Z}/15\mathbb{Z}).$$

For each antidiagonal set $A \subseteq \mathbb{Z}/15\mathbb{Z}$ of the prescribed size a , put

$$p_x = \sum_{y \in A} P_{x,y}, \quad q_x = \sum_{y \notin A} Q_{x,y}, \quad Q_0 = \sum_{x \in \mathbb{Z}/15\mathbb{Z}} q_x.$$

Then for any diagonal set D ,

$$|B(R, C, D, A)| = \sum_{x \in D} p_x, \quad |W(R, C, D, A)| = \sum_{x \notin D} q_x = Q_0 - \sum_{x \in D} q_x.$$

Hence a diagonal set D of the prescribed size d completes the configuration if and only if

$$(6) \quad \sum_{x \in D} p_x \geq 21 \quad \text{and} \quad \sum_{x \in D} q_x \leq Q_0 - 21.$$

This is an exact 15-item, fixed-cardinality, two-constraint subset problem, with nonnegative integer weights $p_x, q_x \leq 15$. Before it is posed, two scalar necessary conditions — $\sum_x p_x \geq 21$ and $Q_0 \geq 21$, both depending on A alone — are tested and cheaply discard most candidates A .

Stage 3: exact completion decision. Problem (6) is decided in two layers, both of which reject only by necessary conditions. First, two cardinality-aware root bounds: let μ be the sum of the d largest of the weights p_x and let λ be the sum of the d smallest of the weights q_x . Every d -element set D satisfies $\sum_{x \in D} p_x \leq \mu$ and $\sum_{x \in D} q_x \geq \lambda$, so if $\mu < 21$ or $\lambda > Q_0 - 21$ the instance is infeasible and is rejected without further work; this layer is where the bulk of the scalar-prefilter survivors die (Table 3). Second, for the instances that pass both root bounds, a depth-first search over the 15 items, each included or excluded, with the following bounds precomputed by suffix dynamic programming: for every position x and every remaining cardinality k , the largest attainable sum of k of the weights $p_x, \dots, p_{14}$, and the smallest attainable sum of k of the weights $q_x, \dots, q_{14}$.

Lemma 14. *The two-layer decision of Stage 3 answers (6) correctly.*

Proof. The root bounds reject only infeasible instances, as just argued. In the search, a node is abandoned only when one of the following holds: the number of remaining items is smaller than the cardinality still required; the accumulated q -sum already exceeds $Q_0 - 21$; the accumulated p -sum plus the maximal attainable remaining p -sum with the required cardinality is below 21; or the accumulated q -sum plus the minimal attainable remaining q -sum with the required cardinality exceeds $Q_0 - 21$. Each is a necessary condition for a feasible completion within the subtree, so no feasible D is discarded. Conversely the search reports success only at a node where the required cardinality is reached and the p -constraint is met, or at a node where the p -constraint is already met and the minimal remaining q -sum with the required cardinality keeps the q -constraint satisfiable; in the latter case an explicit feasible D exists, since p -sums only increase as further items are added. In both cases a feasible D exists. As the search is otherwise a complete binary recursion over the 15 items, it is exhaustive. $\square$

Item order affects only speed, not correctness; the implementation sorts items by a ratio heuristic. Consequently the three stages together decide, for a given profile, whether *any* configuration with that ordered size profile has $|B| \geq 21$ and $|W| \geq 21$, and the answer transfers to the whole D_4 orbit and, in the stratum $S = 30$, to the complementary profile as well.

6. THE SEARCH AND ITS OUTCOME FOR $n = 15$

The following lemma assembles the chain of Sections 2–5 into a single soundness statement for the computation.

Lemma 15 (End-to-end exhaustiveness, $n = 15$). *If a configuration in $\mathbb{Z}/15\mathbb{Z}$ with $|B| \geq 21$ and $|W| \geq 21$ exists, then the search of Section 5, executed on all 247 canonical profiles, reports a feasible completion for at least one of them.*

Proof. By Lemma 6 such a configuration exists with $S \leq 30$. Its size profile satisfies the pair and triple conditions (Lemmas 11 and 9), so by Proposition 12 some element γ of the D_4 action — composed, possibly, with complementation when $S = 30$ — carries the configuration to one whose ordered profile is a canonical worklist profile; γ preserves feasibility. Within that profile’s search, a further symmetry (the choice of fixed pair, Lemma 13, and if applicable the Stage-2 relabelling of the completion families) carries the configuration to one whose fixed pair is an enumerated representative. Its Stage-2 set is among the exhaustively enumerated subsets of the prescribed size, and it passes the scalar conditions of Stage 2, which are necessary. Its remaining line set satisfies (6), so the instance passes the root bounds and, by Lemma 14, the Stage-3 decision reports it feasible. $\square$

Proposition 16. *For every one of the 247 canonical profiles of Proposition 12, no configuration in $\mathbb{Z}/15\mathbb{Z}$ with that profile has $|B| \geq 21$ and $|W| \geq 21$.*

Proof. The search of Section 5 was executed on all 247 profiles by `peace15_solver.cpp` (SHA-256 digest in Section 12) and reported no feasible completion anywhere; by Lemma 15 this establishes the claim. The run statistics are given in Table 3; a per-profile certificate recording, for each profile, the fixed pair, the orbit count, a hash of the representative list, and the case counts, is included among the ancillary files, and the run was independently repeated as described in Section 11.2. $\square$

canonical profiles	247
distinct fixed-pair orbit types	43
Stage-2 completion subsets examined	6,074,753,568
survivors of the scalar prefilter of Stage 2	2,329,772,941
survivors of the root bounds of Stage 3 (searches entered)	26,948,453
branch nodes in those searches	246,476,279
profiles admitting a $21 + 21$ configuration	0

TABLE 3. Statistics of the exhaustive run for $n = 15$, one line per pruning layer: of the 6,074,753,568 enumerated Stage-2 subsets, 2,329,772,941 pass the two scalar conditions of Stage 2, of which 26,948,453 pass the two cardinality-aware root bounds of Stage 3 and enter the depth-first search; every layer rejects by necessary conditions only. The number of Stage-2 subsets examined equals $\sum_{\text{profiles}} (\text{orbit count}) \cdot \binom{15}{a}$, which is checked row by row by the audit script.

Proof of Theorem 1. The lower bound is Proposition 10. For the upper bound, suppose 21 black and 21 white queens are placed peaceably. By Lemma 5 there is a configuration with $|B| \geq 21$ and $|W| \geq 21$, by Lemma 6 one may take $S \leq 30$, and by Proposition 12 its size profile lies in the 247-element canonical list. This contradicts Proposition 16. Hence $t(15) \leq 20$, and $t(15) = 20$. $\square$

Part II: the even case $n = 16$.

7. THE LOWER BOUND FOR $n = 16$

Proposition 17. $t(16) \geq 32$.

Proof. Work in $\mathbb{Z}/16\mathbb{Z}$ and take the black line sets

$$(7) \quad R = C = \{5, 7, 9, 11, 13, 15\}, \quad D = A = \{0, 2, 4, 6, 8, 10, 12, 14\},$$

of sizes $(6, 6, 8, 8)$, and colour every remaining line white. A cell (r, c) is black-homogeneous iff $r, c \in \{5, 7, 9, 11, 13, 15\}$ and $r - c, r + c$ are even; the last two conditions are automatic because r and c are then both odd. Hence

$$B = \{5, 7, 9, 11, 13, 15\} \times \{5, 7, 9, 11, 13, 15\}, \quad |B| = 36.$$

A cell is white-homogeneous iff $r, c \in \{0, 1, 2, 3, 4, 6, 8, 10, 12, 14\}$ and $r - c, r + c$ are both odd, i.e. iff r and c lie in that set and have opposite parity. Writing $E = \{0, 2, 4, 6, 8, 10, 12, 14\}$ and $F = \{1, 3\}$ for its even and odd parts,

$$W = (E \times F) \cup (F \times E), \quad |W| = 8 \cdot 2 + 2 \cdot 8 = 32.$$

By Lemma 5, placing black queens on any 32 of the 36 cells of B and white queens on all 32 cells of W gives a peaceable $32 + 32$ placement. $\square$

Figure 2 displays the configuration, with the four omitted black-homogeneous cells taken to be the corners $(5, 5), (5, 15), (15, 5), (15, 15)$ of the 6×6 block.

This lower bound is not new. It is the $(4, 4)$ -plaid of [2, §3], whose peaceability is their Lemma 3.1, transported to line-set form: under the relabelling $i \mapsto i - 1$ from $\{1, \dots, 16\}$ to $\mathbb{Z}/16\mathbb{Z}$, their white set for $(a, b) = (4, 4)$ is exactly our B and their black set is exactly our W , so that the placement has 32 queens of one colour and 36 of the other, and $\min\{|B|, |W|\} = 32$.

$r \backslash c$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	.	□	.	□	.	.	.	.	.	.	.	.	.	.	.	.
1	□	.	□	.	□	.	□	.	□	.	□	.	□	.	□	.
2	.	□	.	□	.	.	.	.	.	.	.	.	.	.	.	.
3	□	.	□	.	□	.	□	.	□	.	□	.	□	.	□	.
4	.	□	.	□	.	.	.	.	.	.	.	.	.	.	.	.
5	.	.	.	.	.	◇	.	■	.	■	.	■	.	■	.	◇
6	.	□	.	□	.	.	.	.	.	.	.	.	.	.	.	.
7	.	.	.	.	.	■	.	■	.	■	.	■	.	■	.	■
8	.	□	.	□	.	.	.	.	.	.	.	.	.	.	.	.
9	.	.	.	.	.	■	.	■	.	■	.	■	.	■	.	■
10	.	□	.	□	.	.	.	.	.	.	.	.	.	.	.	.
11	.	.	.	.	.	■	.	■	.	■	.	■	.	■	.	■
12	.	□	.	□	.	.	.	.	.	.	.	.	.	.	.	.
13	.	.	.	.	.	■	.	■	.	■	.	■	.	■	.	■
14	.	□	.	□	.	.	.	.	.	.	.	.	.	.	.	.
15	.	.	.	.	.	◇	.	■	.	■	.	■	.	■	.	◇

FIGURE 2. The $32+32$ placement of Proposition 17: ■ black queens, □ white queens, ◇ four of the 36 black-homogeneous cells, left empty. Rows, columns and both diagonal families wrap around.

We reproduce it here in line-set form so that Theorem 2 is self-contained and checkable by hand. Note that Theorem 1.1 of [2] is asymptotic — it asserts $t(n) \geq 0.1339 n^2$ only for sufficiently large even n , and $0.1339 \cdot 256 > 32$ — so it is the construction, not the stated theorem, that supplies the bound at $n = 16$.

8. PARITY-REFINED PROFILES FOR $n = 16$

From now on $n = 16$ and we ask whether a configuration in $\mathbb{Z}/16\mathbb{Z}$ exists with $|B| \geq 33$ and $|W| \geq 33$. By Lemma 6 we may assume $S \leq 32$.

By Lemma 8(ii) the diagonal–antidiagonal incidence is no longer uniform: every cell satisfies $d \equiv a \pmod{2}$, a same-parity pair of labels meets in two cells and an opposite-parity pair in none. The four sizes are therefore not enough data, and we refine them. Write

$$r = |R|, \quad c = |C|, \quad d_0 = |D \cap 2\mathbb{Z}|, \quad d_1 = |D \setminus 2\mathbb{Z}|, \quad a_0 = |A \cap 2\mathbb{Z}|, \quad a_1 = |A \setminus 2\mathbb{Z}|,$$

and put $d = d_0 + d_1$, $a = a_0 + a_1$. The *size profile* is the six-tuple $p = (r, c, d_0, d_1, a_0, a_1)$, with $r, c \in \{0, \dots, 16\}$ and $d_0, d_1, a_0, a_1 \in \{0, \dots, 8\}$, and $S = r + c + d + a$.

Lemma 18 (Pair bounds). *If a configuration in $\mathbb{Z}/16\mathbb{Z}$ with profile p has $|B| \geq 33$ and $|W| \geq 33$, then*

$$(8) \quad rc, rd, ra, cd, ca \geq 33, \quad 2(d_0a_0 + d_1a_1) \geq 33,$$

and the same five products and the same parity-refined expression for the complementary sizes:

$$(9) \quad \begin{aligned} & (16-r)(16-c), (16-r)(16-d), (16-r)(16-a), \\ & (16-c)(16-d), (16-c)(16-a) \geq 33, \\ & 2((8-d_0)(8-a_0) + (8-d_1)(8-a_1)) \geq 33. \end{aligned}$$

Proof. For the five pairs not consisting of the diagonal and antidiagonal families, Lemma 8(i) gives a unique common cell, so distinct cells of B determine distinct pairs of black lines and $|B|$ is at most the product of the two sizes. For the diagonal–antidiagonal pair, Lemma 8(ii) gives two common cells for each same-parity pair of labels and none otherwise, so

$$|B| \leq 2 \# \{(x, y) \in D \times A : x \equiv y \pmod{2}\} = 2(d_0 a_0 + d_1 a_1).$$

The complementary statements are the same bounds applied to (R^c, C^c, D^c, A^c) , whose parity-refined sizes are $(8 - d_0, 8 - d_1, 8 - a_0, 8 - a_1)$ because each parity class of $\mathbb{Z}/16\mathbb{Z}$ has exactly 8 labels. $\square$

Lemma 19 (Triple constraints). *If a configuration in $\mathbb{Z}/16\mathbb{Z}$ with profile p has $|B| \geq 33$ and $|W| \geq 33$, then*

$$(10) \quad 256 - 16(r + c + z) + rc + rz + cz \geq 66 \quad \text{for } z \in \{d, a\},$$

$$(11) \quad 256 - 16(x + d + a) + xd + xa + 2(d_0 a_0 + d_1 a_1) \geq 66 \quad \text{for } x \in \{r, c\}.$$

Proof. Apply Lemma 9 with $n = 16$ and $m = 33$. For the triples $\{R, C, D\}$ and $\{R, C, A\}$ all three pairs are covered by Lemma 8(i), so every $m(X, Y)$ is a product of sizes, giving (10). For the triples $\{R, D, A\}$ and $\{C, D, A\}$ the diagonal–antidiagonal term is $2(d_0 a_0 + d_1 a_1)$ by Lemma 8(ii) and the other two are products, giving (11). $\square$

Symmetry. Three involutions act on profiles, each induced by an affine symmetry of the board:

$$\begin{aligned} \tau(r, c) = (c, r) : & \quad R \leftrightarrow C, \quad D \mapsto -D, \quad A \mapsto A; \\ \nu(r, c) = (r, -c) : & \quad R \mapsto R, \quad C \mapsto -C, \quad D \leftrightarrow A; \\ \sigma(r, c) = (r + 1, c) : & \quad R \mapsto R + 1, \quad C \mapsto C, \quad D \mapsto D + 1, \quad A \mapsto A + 1. \end{aligned}$$

Negation and translation by an even amount preserve the parity of a line label, so τ acts on profiles by $r \leftrightarrow c$ and ν by $(d_0, d_1) \leftrightarrow (a_0, a_1)$. Translation by an odd amount in one coordinate shifts both the diagonal and the antidiagonal label by an odd amount, so σ acts by the *simultaneous* exchange $d_0 \leftrightarrow d_1, a_0 \leftrightarrow a_1$. Unlike the odd case there is no halving map: 2 is not a unit modulo 16, the matrix $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ has determinant -2 and is not invertible over $\mathbb{Z}/16\mathbb{Z}$, and indeed the two pairs of families are not interchangeable, since a row and a column always meet in one cell while a diagonal and an antidiagonal meet in two or in none. The group generated by the three profile involutions has order at most 8.

Proposition 20. *Let (R, C, D, A) be a configuration in $\mathbb{Z}/16\mathbb{Z}$ with $|B| \geq 33$, $|W| \geq 33$ and $S \leq 32$. Then its profile is one of exactly 1,898 ordered six-tuples satisfying (8), (9), (10), (11) and $S \leq 32$, falling into exactly 342 classes under the group generated by τ, ν, σ , distributed by S as in Table 4.*

S	24	25	26	27	28	29
canonical profiles	13	41	101	120	57	10

TABLE 4. The 342 canonical parity-refined profiles for $n = 16$, by budget S . No profile with $S \leq 23$ or $S \geq 30$ survives the necessary conditions.

Proof. Enumerate all $17^2 \cdot 9^4 = 1,896,129$ six-tuples, filter by $S \leq 32$ and by the conditions of Lemmas 18 and 19, which are necessary, and canonicalize the survivors by taking the lexicographically least element of the orbit under $\langle \tau, \nu, \sigma \rangle$. The computation is finite and was carried out independently twice, in C++ and in Python, giving 1,898 ordered and 342 canonical profiles with the distribution of Table 4. $\square$

Remark 21. Table 4 is a structural fact worth isolating: the necessary conditions alone already exclude every profile with $S \geq 30$, so none of the top three strata $S \in \{30, 31, 32\}$ permitted by Lemma 6 needs to be searched at all. Contrast the odd case, where the top stratum $S = 30$ contains 39 of the 247 canonical profiles. No single one of the conditions of Lemmas 18 and 19 is responsible — the exclusion is a joint effect and is part of the finite enumeration — but the balanced profile at the top of the budget shows what is happening. For n even and $x = y = z = n/2$, the identity (2) applied to $\{R, C, D\}$ gives exactly

$$T + T' = n^2 - n \cdot \frac{3n}{2} + 3 \cdot \frac{n^2}{4} = \frac{n^2}{4},$$

whereas $|B| + |W| \geq 2m$ with $m = n^2/8 + 1$ demands $n^2/4 + 2$. The balanced top-stratum profile misses by exactly 2, for every even n ; at $n = 16$ this reads $64 < 66$.

9. DECIDING ONE PROFILE EXACTLY FOR $n = 16$

Fix a canonical profile. Since the halving map is unavailable, the fixed pair is always (R, C) ; we may assume $r \leq c$ after a transpose. The search again proceeds in three stages.

Stage 1: representatives of the row–column pair, and parity orientation. Fix the pair (R, C) up to the group generated by

- independent translations $(R, C) \mapsto (R + \alpha, C + \beta)$ for $\alpha, \beta \in \mathbb{Z}/16\mathbb{Z}$ (256 elements);
- common unit scaling $(R, C) \mapsto (uR, uC)$ for odd u (8 elements), which scales D and A by the same odd u and hence preserves all four parity-refined sizes;
- the transpose τ , admissible exactly when $r = c$;
- the relative sign ν , i.e. $(R, C) \mapsto (R, -C)$, admissible *only* when $(d_0, d_1) = (a_0, a_1)$, since ν exchanges the diagonal and antidiagonal families.

The group has order $256 \cdot 8 = 2048$, doubled for each admissible extra generator.

Translations require the same care as the relative sign, and for the same reason. A translation with $\alpha - \beta$ odd sends $D \mapsto D + (\alpha - \beta)$ and $A \mapsto A + (\alpha + \beta)$ with both shifts odd, hence flips the two parity splits simultaneously: it maps the ordered profile $(r, c, d_0, d_1, a_0, a_1)$ to $(r, c, d_1, d_0, a_1, a_0)$, a different ordered profile in general. Quotienting the row–column pair by *all* 256 translations is therefore sound only if both parity orientations of every canonical profile are searched. They are: each canonical profile is expanded into the set $\{p, \sigma p\}$ of oriented profiles, which has two elements unless p is σ -invariant. Across the 342 canonical profiles this produces 677 oriented jobs (335 profiles contribute two and 7 contribute one).

Within one oriented job, the diagonal and antidiagonal families may still be exchanged, because ν does so globally, so the ordered profiles $(r, c, d_0, d_1, a_0, a_1)$ and $(r, c, a_0, a_1, d_0, d_1)$ are simultaneously feasible or infeasible; the implementation puts the family with the smaller number of admissible masks in the role of A , which is the family enumerated in Stage 2. The transfer argument of Lemma 13 applies verbatim to the group above, with its admissibility conditions (τ only when $r = c$, ν only when $(d_0, d_1) = (a_0, a_1)$): soundness requires only that the enumerated pairs meet every orbit, and the emitted set is produced by the same necklace sweep with explicit orbit marking as in the odd case. Only 14 distinct triples

$(r, c, \text{sign admissible})$ occur across the 677 jobs, and for each of them the cardinality of the emitted set equals the orbit count computed independently by Burnside's lemma (Section 11).

Stage 2: the completion as a parity-constrained subset problem. Fix a representative (R, C) . For labels $x, y \in \mathbb{Z}/16\mathbb{Z}$ define

$$P_{x,y} = \#\{(r, c) : r - c = x, r + c = y, r \in R, c \in C\},$$

$$Q_{x,y} = \#\{(r, c) : r - c = x, r + c = y, r \notin R, c \notin C\}.$$

By Lemma 8(ii) these are 0 when $x \not\equiv y \pmod{2}$ and lie in $\{0, 1, 2\}$ otherwise, the two candidate cells being (r, c) and $(r + 8, c + 8)$ for the solution $r = \lfloor ((y + x) \bmod 16)/2 \rfloor$, $c = \lfloor ((y - x) \bmod 16)/2 \rfloor$ or its shift by 8 in one coordinate, whichever satisfies the two original congruences.

For each antidiagonal set $A \subseteq \mathbb{Z}/16\mathbb{Z}$ with $|A \cap 2\mathbb{Z}| = a_0$ and $|A \setminus 2\mathbb{Z}| = a_1$, put

$$p_x = \sum_{y \in A} P_{x,y}, \quad q_x = \sum_{y \notin A} Q_{x,y}, \quad Q_0 = \sum_{x \in \mathbb{Z}/16\mathbb{Z}} q_x,$$

so that $|B(R, C, D, A)| = \sum_{x \in D} p_x$ and $|W(R, C, D, A)| = Q_0 - \sum_{x \in D} q_x$ for every D . Hence a diagonal set D with $|D \cap 2\mathbb{Z}| = d_0$ and $|D \setminus 2\mathbb{Z}| = d_1$ completes the configuration if and only if

$$(12) \quad \sum_{x \in D} p_x \geq 33 \quad \text{and} \quad \sum_{x \in D} q_x \leq Q_0 - 33.$$

Before (12) is posed, the necessary conditions $\sum_x p_x \geq 33$ and $Q_0 \geq 33$ are tested; they discard all but a vanishing fraction of the candidates A (Table 5).

Stage 3: exact completion decision. Problem (12) is a 16-item, two-constraint subset problem with a prescribed cardinality *in each parity class*. It is decided in two ways, and both were run over the whole search (Section 11).

The first is a parity-aware extremal bound followed, if the bound is inconclusive, by a depth-first search with suffix dynamic-programming bounds indexed by the two remaining cardinalities:

Lemma 22. *Let μ be the sum of the d_0 largest values p_x over even x plus the d_1 largest over odd x , and let λ be the sum of the d_0 smallest values q_x over even x plus the d_1 smallest over odd x . If $\mu < 33$ or $\lambda > Q_0 - 33$, then (12) is infeasible. Otherwise the depth-first search decides (12) correctly.*

Proof. Any admissible D selects exactly d_0 even and d_1 odd labels, so $\sum_{x \in D} p_x \leq \mu$ and $\sum_{x \in D} q_x \geq \lambda$; each of the two displayed inequalities therefore contradicts one of the two conditions in (12). For the search, a node is abandoned only when one of the following holds: the remaining cardinality required in some parity class is negative or exceeds the number of remaining labels in that class; the accumulated q -sum already exceeds $Q_0 - 33$; the accumulated p -sum plus the maximal attainable remaining p -sum with the required parity-split cardinality is below 33; or the accumulated q -sum plus the minimal attainable remaining q -sum with that cardinality exceeds $Q_0 - 33$. Each is a necessary condition for a feasible completion within the subtree, so no feasible D is discarded. Success is reported only when the required cardinalities are reached and the p -constraint is met, or when the p -constraint is already met at a node whose minimal remaining q -sum keeps the q -constraint satisfiable; in both cases a feasible D exists, since p -sums only increase as further labels are added. The search is otherwise a complete binary recursion over the 16 labels. $\square$

The second is direct enumeration: there are exactly $\binom{8}{d_0}\binom{8}{d_1} \leq \binom{8}{4}^2 = 4,900$ admissible masks D , and each is tested against (12). This is obviously exhaustive and involves no bound and no pruning at all.

Consequently the three stages together decide, for a given oriented profile, whether *any* configuration with that profile has $|B| \geq 33$ and $|W| \geq 33$, and the answer transfers to the whole orbit under $\langle \tau, \nu, \sigma \rangle$.

10. THE SEARCH AND ITS OUTCOME FOR $n = 16$

Lemma 23 (End-to-end exhaustiveness, $n = 16$). *If a configuration in $\mathbb{Z}/16\mathbb{Z}$ with $|B| \geq 33$ and $|W| \geq 33$ exists, then the search of Section 9, executed on all 677 oriented jobs, reports a feasible completion for at least one of them.*

Proof. By Lemma 6 such a configuration exists with $S \leq 32$. Its parity-refined profile satisfies the conditions of Lemmas 18 and 19, so by Proposition 20 some element of $\langle \tau, \nu, \sigma \rangle$ carries it to a configuration whose ordered profile is one of the 677 oriented profiles searched; each generator is induced by a board symmetry and preserves feasibility. Within that job, a symmetry of the Stage-1 group (with the transfer argument of Lemma 13) and, if applicable, the Stage-2 relabelling by ν carry the configuration to one whose row-column pair is an enumerated representative. Its antidiagonal set is among the exhaustively enumerated admissible masks, and it passes the two scalar conditions of Stage 2, which are necessary. Its diagonal set satisfies (12), so by Lemma 22 — or trivially by the brute-force kernel — the Stage-3 decision reports it feasible. $\square$

Proposition 24. *For every one of the 677 oriented profiles derived from the 342 canonical profiles of Proposition 20, no configuration in $\mathbb{Z}/16\mathbb{Z}$ with that profile has $|B| \geq 33$ and $|W| \geq 33$.*

Proof. The search of Section 9 was executed twice on all 677 jobs, by `peace16_solver.cpp` and its brute-force variant (SHA-256 digests in Section 12), and reported no feasible completion anywhere; by Lemma 23 this establishes the claim. The run statistics are given in Table 5; a per-job certificate recording the oriented profile, the pair-orbit count, the number of admissible antidiagonal masks and the case counts is included among the ancillary files, for each of the two completion kernels. $\square$

The collapse from $1.3 \cdot 10^{10}$ candidates to 17,834 is caused entirely by the two necessary conditions $\sum_x p_x \geq 33$ and $Q_0 \geq 33$ of Stage 2; both are conditions on the antidiagonal set alone, and on the even torus they are severe.

Proof of Theorem 2. The lower bound is Proposition 17. For the upper bound, suppose 33 black and 33 white queens are placed peaceably. By Lemma 5 there is a configuration with $|B| \geq 33$ and $|W| \geq 33$, by Lemma 6 one may take $S \leq 32$, and by Proposition 20 its parity-refined profile lies in the 342-element canonical list, hence its ordered profile is one of the 677 oriented profiles searched. This contradicts Proposition 24. Hence $t(16) \leq 32$, and $t(16) = 32$. $\square$

11. VERIFICATION

An exhaustive computation is only as good as the evidence that it is exhaustive and correctly implemented. We list the checks performed, case by case. The evidence is not equally strong on the two sides, and the difference is stated plainly in Section 11.3.

canonical profiles	342
oriented profile jobs	677
distinct row-column pair orbit types	14
Stage-2 completion subsets examined	13,163,028,768
survivors of the scalar prefilter of Stage 2	17,834
survivors of the root bounds of Stage 3 (searches entered)	0
branch nodes in the depth-first search	0
jobs admitting a $33 + 33$ configuration	0

TABLE 5. Statistics of the exhaustive run for $n = 16$, one line per pruning layer as in Table 3. The number of Stage-2 subsets examined equals $\sum_{\text{jobs}}(\text{pair-orbit count}) \cdot \binom{8}{a_0} \binom{8}{a_1}$, which is checked row by row by the audit script. The parity-aware root bound of Lemma 22 already refuted each of the 17,834 scalar-prefilter survivors, so the depth-first search was never entered; the same 17,834 instances were also refuted by direct enumeration of all admissible masks by the brute-force kernel.

11.1. What both cases have in common. *Startup gates.* Each solver begins with gates that abort the run on failure rather than proceeding: a randomized comparison of its exact completion decision against direct enumeration of all admissible masks, a check of known orbit counts, and a regeneration of the profile list from the necessary conditions.

Memory safety. AddressSanitizer and UndefinedBehaviorSanitizer builds of both proof solvers pass all startup self-tests; beyond that, an ASan+UBSan build of each proof solver re-ran its *complete* sweep — all 247 profiles for $n = 15$ (232 s single-threaded) and all 677 jobs for $n = 16$ (244 s) — with zero sanitizer diagnostics and certificates identical to the recorded runs in every non-timing column.

Independent audit scripts. A Python audit script for each case, sharing no code with the solvers, re-derives the profile list, recomputes every fixed-pair orbit count from Burnside’s lemma — counting, for each group element, the subsets of each cardinality invariant under the induced affine permutation of $\mathbb{Z}/n\mathbb{Z}$, via the cycle structure — checks the per-row and total case arithmetic of the certificates, and re-counts the lower-bound witness. Both print `AUDIT_OK`.

What is not claimed. The certificate files are audit logs, not standalone proof objects in the sense of DRAT or LRAT: they record what was searched, not a machine-checkable refutation. The refutation is the short program together with the reductions of Sections 2–9; re-running the program is what re-establishes it, and agreement between implementations is a strong error filter, not a formal guarantee.

11.2. The case $n = 15$. The two enumerators mentioned below — the *proof solver* and the *independent enumerator* — were written separately, from the mathematical description of Section 5, in different C++ dialects, by different systems (see Section 13), and share no code.

Two independent enumerators. The proof solver (`peace15_solver.cpp`) and the independent enumerator (`profile_enum.cpp`) agree that all 247 profiles are infeasible. They differ in three substantive respects. First, in the symmetry quotient: the two select different fixed pairs on 19 of the 247 profiles, and consequently enumerate different case sets — 6,074,753,568 antidiagonal subsets for the solver against 6,241,793,402 completions for the enumerator. Second, in the decision procedure for (6): the solver uses the depth-first

search with suffix dynamic-programming bounds of Stage 3, whereas the enumerator uses an exact $7 + 8$ meet-in-the-middle table, retaining for each cardinality and each p -sum threshold the minimum achievable q -sum over right-half subsets — an algorithm with no search and no pruning at all. Third, in scheduling: the solver is OpenMP-parallel, the enumerator single-threaded and resumable, writing one JSON record per profile.

Agreement on the profile list. The 247-profile worklist was generated independently by us in Python, from Lemmas 11 and 9; the proof solver regenerates the set from scratch in C++ at startup and aborts unless it coincides with the supplied file, which it does byte for byte; the audit script regenerates it a third time in Python. At $S = 30$ the count was also reproduced by an independent hand-checkable enumeration as 370 ordered profiles falling into 39 canonical classes.

Repeated runs on different toolchains. The proof solver was run three times: twice as recorded by its author (five threads, 60.4 s; three threads, 74.0 s) and once by us from a fresh build with a different compiler and no OpenMP (single-threaded, 101.7 s). All 247 rows of the certificate agree in every column except the per-profile timing.

Brute-force gates on the completion kernel. At startup the proof solver compares its Stage 3 decision against direct enumeration of all $\binom{15}{k}$ subsets on 20,000 deterministic pseudorandom instances with randomized weights and cardinalities, and aborts on any disagreement; the enumerator runs the same style of gate on 100,000 instances. The solver additionally checks three known pair-orbit counts before starting: 11,793 for $(r, c) = (7, 7)$ under the order-3600 group (transpose, no relative sign), 6,892 for $(7, 7)$ under the order-7200 group (transpose and relative sign), and 13,654 for $(7, 8)$ with the relative sign; the first two had been derived independently earlier in the project, by Burnside’s lemma.

Burnside audit. The audit script (`peace15_audit.py`) recomputes all 43 fixed-pair orbit counts from Burnside’s lemma and compares them with the counts recorded in the certificate. It also verifies that each certificate row satisfies $\text{cases} = \text{orbits} \times \binom{15}{a}$, verifies the column total 6,074,753,568, and re-counts the witness of Proposition 10 as 20 black-homogeneous and 22 white-homogeneous cells.

The witness. The $20 + 20$ placement of Proposition 10 was checked by four independently written scripts, including a from-scratch check that verifies all 400 black–white pairs against all four attack relations directly, rather than through the line-set formulation. That check is included among the ancillary files as `check_witness15.py`; the other three (the audit script’s recount, and two checks in the project repository) are described here for completeness.

Source audit. The proof solver was read line by line against the mathematics of Section 5, with particular attention to the three places where an unsound shortcut would be invisible in the output: the admissibility conditions for the transpose and relative-sign quotients, the necessary-condition character of every pruning rule, and the reconstruction path that would turn a reported feasible instance into an explicit quadruple verified cell by cell.

Independent partial evidence. A SAT-based cube-and-conquer campaign on a 3,244-variable encoding of the $21 + 21$ question, later superseded and never finished, produced roughly 28,900 DRAT-certified cube refutations checked by `drat-trim` in default backward mode; none of the processed cubes was satisfiable, though the cube partition was never completed. It proves nothing on its own but is consistent with Theorem 1. A per-profile certified-CNF pilot refuted $(5, 5, 5, 5)$ with a checked 9.2 MB DRAT proof but timed out on $(6, 8, 7, 7)$ under two solvers; this is why an exhaustive enumeration, and not a proof-logging pipeline, carries the upper bound.

11.3. The case $n = 16$. The gap in the evidence, stated first. For $n = 16$ there is as yet no fully independent second enumerator. Two programs were written and run to completion, `peace16_solver.cpp` and `peace16_solver_bruteforce.cpp`, but they are the same program apart from the completion kernel: profile generation, the symmetry quotient, the pair-representative construction, the parity orientation logic and the antidiagonal loop are shared verbatim, and the two files differ in ten lines. A systematic error in the shared part — for instance an unsound quotient that discarded a feasible configuration before the kernel ever saw it — would not be caught by their agreement. This is a real weakness relative to the $n = 15$ case, and closing it by writing a genuinely independent enumerator from Sections 8–9 is the single most valuable check that remains open. What follows should be read against that reservation.

Two completion kernels over the whole search. The two programs agree on every non-timing column of all 677 certificate rows: the same 13,163,028,768 antidiagonal subsets examined, the same 17,834 survivors of the prefilter, and UNSAT for every job. Where they differ, they differ substantively. The optimized kernel refuted each of the 17,834 surviving instances by the parity-aware extremal bound of Lemma 22 alone, without entering its depth-first search — the certificates record 0 branch nodes. The brute-force kernel refuted the same 17,834 instances by testing every one of the $\binom{8}{d_0}\binom{8}{d_1} \leq 4,900$ admissible masks. So the instances that actually decide the theorem were settled twice, once by a counting bound and once by complete enumeration, with no shared code on that path. Each binary independently passes a startup gate comparing the two kernels on 20,000 deterministic pseudorandom instances with randomized weights and parity cardinalities.

Independent audit. The audit script `peace16_audit.py` shares no code with the solvers and re-derives, in Python: the 1,898 ordered and 342 canonical profiles and their strata from the necessary conditions of Lemmas 18 and 19; all 14 row–column pair-orbit counts by Burnside’s lemma; the two-lift reconstruction of Lemma 8(ii), by comparing the closed-form formula used in the solver against brute-force enumeration of the common cells for all 256 label pairs (d, a) ; the lower-bound witness (7), recounted as 36 black-homogeneous and 32 white-homogeneous cells; the row arithmetic and the column totals 13,163,028,768 and 17,834 of both certificates, and their column-by-column agreement. It was re-run locally during preparation of this note and prints `AUDIT_OK`.

Startup gates in the solvers. Beyond the 20,000-instance kernel comparison, each solver checks that the number of 8-element necklaces in $\mathbb{Z}/16\mathbb{Z}$ is 810 and that two known pair-orbit counts, 73,663 for $(r, c) = (7, 8)$ and 42,062 for $(8, 8)$ without the relative sign, come out right; it regenerates the profile list and aborts unless it finds 1,898 ordered and 342 canonical profiles. Had any job been satisfiable, the solver would have reconstructed an explicit quadruple (R, C, D, A) and verified it by direct enumeration of the 256 cells before reporting; that path is exercised by the self-tests and was never taken in the run.

Local rebuild and rerun. During preparation of this note both programs were additionally rebuilt with a different toolchain (Apple `clang` on `arm64`, no OpenMP, single-threaded) and rerun to completion; all 677 rows of both regenerated certificates agree with the recorded five-thread runs in every non-timing column. The startup gates of the preceding paragraph were exercised by these reruns.

The witness. The $32 + 32$ placement of Proposition 17 was verified independently of the line-set formulation by checking all $32 \cdot 32 = 1024$ black–white pairs against all four attack relations directly (included among the ancillary files as `check_witness16.py`), and separately

by reconstructing the $(4, 4)$ -plaid from the definition in [2, §3] and confirming that it coincides cell for cell with (7) after the relabelling $i \mapsto i - 1$.

Source audit. The proof solver was read against the mathematics of Sections 8–9, with particular attention to the two soundness points peculiar to the even case: that quotienting the row–column pair by all 256 translations is compensated by searching both parity orientations of every canonical profile, and that the relative-sign quotient is applied only when $(d_0, d_1) = (a_0, a_1)$.

12. REPRODUCTION AND ANCILLARY FILES

The ancillary files accompanying this note are:

$n = 15$	
peace15_solver.cpp	the proof solver (C++17, OpenMP)
profile_enum.cpp	the independent enumerator (C++20)
peace15_audit.py	profile, Burnside and witness audit (Python 3)
WORKLIST-2026-07-25.tsv	the 247 canonical profiles
peace15_certificate.tsv	per-profile certificate, five-thread run
peace15_certificate_run2.tsv	per-profile certificate, three-thread run
$n = 16$	
peace16_solver.cpp	the proof solver (C++20, OpenMP)
peace16_solver_bruteforce.cpp	the same, with the brute-force kernel
peace16_audit.py	profile, Burnside, lift-map and witness audit
peace16_certificate.tsv	per-job certificate, optimized kernel
peace16_certificate_bruteforce.tsv	per-job certificate, brute-force kernel
check_witness15.py	pairwise check of the $20 + 20$ witness
check_witness16.py	pairwise check of the $32 + 32$ witness
README.txt	instructions, certificate schema, expected output
SHA256SUMS	digests of every other ancillary file

To rebuild the upper bound for $n = 15$ (portable verification build):

```
g++ -O3 -fopenmp -std=c++17 peace15_solver.cpp -o peace15_solver
OMP_NUM_THREADS=5 ./peace15_solver \
  --worklist WORKLIST-2026-07-25.tsv \
  --output peace15_certificate.tsv
python3 peace15_audit.py
```

The solver prints SELFTEST_OK after its startup gates and ALL_UNSAT on completion; the audit script prints AUDIT_OK. The recorded five-thread run took about 60 seconds. The independent enumerator is built and run by

```
clang++ -O3 -std=c++20 profile_enum.cpp -o profile_enum
./profile_enum --self-test --random 100000
./profile_enum --batch --worklist WORKLIST-2026-07-25.tsv \
  --results results.jsonl --log logs
```

and takes about 200 seconds single-threaded.

To rebuild the upper bound for $n = 16$ (portable verification build):

```
g++ -O3 -fopenmp -std=c++20 peace16_solver.cpp -o peace16_solver
OMP_NUM_THREADS=5 ./peace16_solver --output peace16_certificate.tsv
```

```
g++ -O3 -fopenmp -std=c++20 \
    peace16_solver_bruteforce.cpp -o peace16_solver_bruteforce
OMP_NUM_THREADS=5 ./peace16_solver_bruteforce \
    --output peace16_certificate_bruteforce.tsv
```

```
python3 peace16_audit.py
```

Each solver prints `SELFTEST_OK` after its startup gates and `ALL_UNSAT` on completion; the audit script prints `AUDIT_OK` and compares the two certificates column by column. The recorded five-thread runs took about 78 seconds each. Runtime is not part of either proof. Note that both audit scripts read the certificates from their own directory, so run them after the solvers have written those files, or against the shipped copies.

Three remarks on the build and run environment. First, `-march=native` and `-DNDEBUG` may be added for speed and were used in the recorded runs; they are omitted from the portable commands above because neither affects any verification step — in particular, no proof-relevant check in any of the programs is implemented via `assert` (the only assertion is an internal size-ordering invariant), all gates being explicit runtime tests that abort the program and are active in every build. Second, the randomized self-tests are deterministic: they use `std::mt19937_64` with the fixed seeds `0x4d595df4d0f33173` ($n = 15$, 20,000 instances) and `0x16e0ddc0ffee` ($n = 16$, 20,000 instances per solver), so every rerun tests the same instances. Third, the recorded five-thread runs were produced with GCC/OpenMP in an x86-64 (AMD EPYC) container, and the independent local reruns — serial, and under AddressSanitizer/UndefinedBehaviorSanitizer — with Apple `clang` 21 on an `arm64` (Apple Silicon) machine running macOS; agreement of all non-timing certificate columns across these environments is part of the evidence of Section 11.

SHA-256 digests of the files that carry the two upper bounds, as recorded:

```
peace15_solver.cpp
1ae5b340d6feb4591dae4f7b57a283a373ce60fd2bc9767fc043f25d402bebe9

peace15_certificate.tsv
f9d2c858cc6e3a144a10da509a744114080de801df6d216b8cd2783cb2f410b0

peace16_solver.cpp
ff8de5710446f1b8721e372a3f9af70bcd05b5798cb7924bce58d7e9c784df7c

peace16_solver_bruteforce.cpp
4153d2181642b3de83f3f0f366d85b8438350aa3b620580cc3509faf7c65c3f

peace16_certificate.tsv
850fc499127193539c06216aa57e91fd6373b3ea2773f4c642654e95def0c086

peace16_certificate_bruteforce.tsv
8f606eac6e0b3bcc31f07b21bcfabf1fddd5d6abd350c9b557765f01dc2948d2
```

13. USE OF AI TOOLS

This note was produced with substantial use of AI systems, which we disclose in the interest of transparency. The reduction programme — the line-colouring equivalence, the budget inequality, the pair and triple bounds, the symmetry analysis and the 247-profile worklist — was developed inside an AI-assisted research repository in which Anthropic Claude agents ran the computations and maintained the standing documents, and in which a large-language-model collaborator acting as a mathematician was consulted in writing; the exact completion

algorithm of Section 5 originates in that consultation. One error in that reply (a claim that the budget could be normalized to $S = 2n$) was found and corrected here; see Remark 7. The final proof documents for both cases, all three proof solvers and both audit scripts listed in Section 12 were produced by OpenAI GPT-5.6 Pro on July 25, 2026.

The verification described in Section 11 was carried out by Anthropic Claude agents: for $n = 15$, the second exact enumerator `profile_enum.cpp` and its brute-force gates, the line-by-line audit of the proof solver, the local rebuild and rerun on a different toolchain, the sanitizer builds, the re-derivation and numerical checking of every identity in Sections 2–4, and the independent checks of the witness; for $n = 16$, the re-derivation of the identities of Sections 8–9, the source audit of the solver, the independent recount of the witness and of the (4, 4)-plaid, and the local re-run of the audit script. The soundness correction to the fixed-pair quotient described in Section 5 was found during that verification.

Agreement between AI-written implementations is a useful and, for $n = 15$, deliberately engineered error filter, but it is not independent verification in the scholarly sense, and it does not replace the proofs given in the text. For $n = 16$ even that filter is weaker, as Section 11.3 states. Readers who wish to check the theorems should re-run the ancillary programs or, better, write a third implementation from Sections 2–9. The author reviewed the manuscript and takes full responsibility for its content.

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